

98 @ The Office



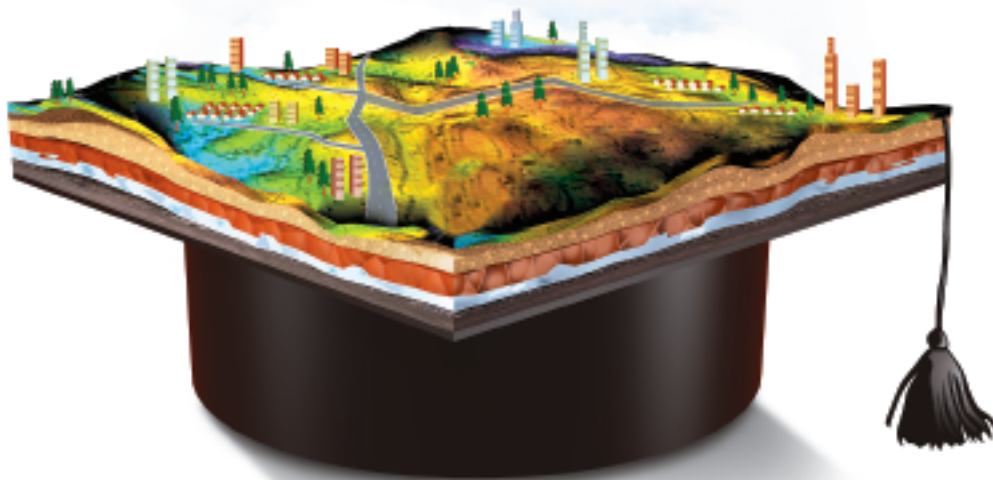
102 Biometrics for Security



104 Number Game

Digital Business

How Technology Can Help Your Organisation



Tech Careers

Mapping A Future

Fascinated by technologies like GPS? Here's how to make it a career!

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It's important to know where you are in the world; this is why we have Global Positioning Satellite (GPS) even entering our daily lives—the latest high-end mobile phones, cameras and who knows what else. Industries like logistics have depended on it to get to their destination on time. But where does all the information on your GPS device come from?

GPS is part of a much larger umbrella called Geographical Information Systems (GIS). By definition, Geographical Information Systems are computer-based tools and technologies used to collect, convert, analyse, model, produce and demonstrate spatial information. Simply put, it's all the information from your geography textbook put into a database. Today GIS is used in natural resource management, environmental planning, disaster management, defence, transport and logistics, urban development, public utilities works, and business applications where you need to know the geography of the area.

I'm Special

Let's start with what you can specialise in:
GPS: Global Positioning Satellite systems do just

that—use a satellite to tell you your position on the Earth. This is easily the most widely used GIS, and is a great enabler for many more.

Remote Sensing: Remote sensing is the science of obtaining information about locations on Earth—its natural resources, for example—without physically going to the location. The Earth's surface is scanned by using the sensors on satellites, and can then be analysed using electromagnetic radiation.

Photogrammetry: Photogrammetry is the technique of creating 2D or 3D models, images, maps or scale drawings of a region, based on photographs. Put simpler, it's the technique of telling how high a mountain is by clicking a couple of pictures from different angles and making a few calculations. The art is incredibly old; today, Microsoft's PhotoSynth represents its future.

Spatial Image Processing And Analysis: Spatial means 'pertaining to space'. Digital images from satellites are analysed to create data structures, and then analysed mathematically. This also involves conversion (from analogue to digital and so on), enhancement, display and finally drawing information from those images.

Projection System: The method of transforming real-life surfaces to two-dimensional maps is called a Projection System. This process requires